UNIT 6 INFLATION AND UNEMPLOYMENT *

Structure

6.0 OBJECTIVES

After going through this unit you should be able to

- identify various types of unemployment;
- explain the concept of natural rate of unemployment;
- establish a relationship between unemployment and inflation;
- explain how the short-run Phillips curve shifts; and
- reconcile the difference in shape of the Phillips curve in short-run and long-run.

6.1 INTRODUCTION

In Units 7 and 8 of BECC 103 we gave some preliminary ideas of inflation – its definition, causes and effects. In this Unit we describe the relationship between inflation and unemployment. In the process, we discuss various types of unemployment and its measurement. Recall that classical economists believed in dichotomy of real and monetary variables. Thus, inflation being a monetary variable should not have any effect on a real variable such as unemployment. Keynesian economists, however, believed that change in monetary variables could affect real variables.

Inflation, as you know, is defined as a *persistent* rise or, a tendency towards persistent rise in the *general level of prices*. As and when there will be

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increase in the general price level, purchasing power of households decline. Increase in inflation is likely to reduce the value of national currency and it's vice versa. Although there are many types of inflation, it can be mainly classified into three types of inflation- demand pull (caused by increase in demand), cost push (cause by increase in cost of production) and built-in inflation (mainly caused by past events). The rate of inflation could vary from a low level to a very high level. As of 2021, Venezuela for example has an inflation rate of nearly 10,000 per cent per year.

Unemployment is a state in which healthy person fails to get employment at prevailing wage. It is caused by various factors that are concerned with demand and supply. There are broadly six types of unemployment- structural (arises due to change in structure of economy, frictional (arises due to time gap between changing jobs), cyclical (arises due to change in cycle of economy i.e. boom, recession, etc.), seasonal (arises due to change in season), disguised unemployment (it is also called hidden unemployment where marginal product of labour is very much close to zero) and under employment in which better qualified person end with low quality, low paid jobs because of excess supply of labour and lower availability of job opportunity.

6.2 TYPES OF UNEMPLOYMENT

'Labour force' as a concept includes all persons in the age group of 16 years to 64 years who are willing to work. Thus it includes both employed and unemployed persons. The persons not included in the labour force include those who are retired, too ill to work, keeping the house, or simply not looking for work.

'Work force' as a concept is somewhat narrower – it includes the employed persons only. Thus the difference between the labour force and the work force gives us the number of unemployed.

By employed persons we mean those who perform any paid work (thus homemakers are not included) and those who have jobs. On the other hand, the unemployed as a category includes people who are not employed but are actively looking for work. While considering unemployment we do not take into account those who are not in the labour force. We define unemployment rate as the number of unemployed divided by the total labour force. You should remember that the concept of unemployment implies 'involuntary unemployment'. This concept implies that a person is willing to work at the prevailing wage rate, but cannot find work.

There are three types of unemployment, viz., frictional, structural and cyclical. We explain the differences below.

(i) Frictional unemployment: It takes place because people switch over from one job to another. In many cases the tenure of job gets over and workers remain unemployed till they get another job. In other cases workers migrate from one region to another in search of better jobs or opt to remain out of job for short time periods. Frictional unemployment takes place because in an economy with imperfect information, job search and matching is not smooth and there are frictions in the economy.

(ii) Structural unemployment: It results from the mismatch between supply and demand for different kinds of jobs. For example, in recent years, the number of engineers and management professionals looking for jobs in India has been much higher than available jobs. This has resulted in a number of persons with technical qualification opting for low qualification jobs. Structural unemployment takes place largely due to structural shifts in an economy and adjustments to such shifts take time. A large number of educational institutions in India have discontinued their engineering education programmes.

(iii) Cyclical unemployment: It arises due to fluctuations in aggregate demand, which is a part of business cycles. When aggregate demand declines, there is simultaneous decline in the demand for labour and consequent increase in unemployment. On the other hand, a general boom in the economy increases the demand for labour and unemployment decreases. Thus cyclical unemployment is pro-cyclical in nature.

Empirical data shows that the labour force in an economy is much less than the total population. Total labour force in India, according to certain sources, is about 50 crores compared to an estimated population of 138 crores in 2020. Persons above 65 years and children below 15 years of age however should not be taken into consideration while comparing the size of the labour force to total population. A relevant ratio in this context is the 'Labour Force Participation Rate (LFPR)'. It is defined as follows:

$$\text{LFPR} = \frac{\text{Size of the labour force}}{\text{Size of population in the age group of } 16 - 64 \text{ years}}$$

The labour force participation rate (LFPR) varies across countries, and over time for the same country. If we take gender into account, there could be male labour force participation rate and female labour force participation rate. Usually, there is a gap between male LFPR and female LFPR. In India, for example, female LFPR is much lower compared to male LFPR. Further, there is a sharp decline in female LFPR in recent years. Such a decline could be due to cultural and structural issues.

The rate of unemployment u is defined as the ratio of unemployed persons to total labour force. The rate of unemployment varies across countries and for a country over time.

6.3 PHILLIPS CURVE

The Phillips curve, named after A W Phillips, describes the relationship between unemployment and inflation. In 1958 Phillips, then professor at London School of Economics, took time series data on the rate of unemployment and the rate of increase in nominal wage rate for the United Kingdom for the period 1861-1957 and attempted to establish a relationship. He took a simple linear equation of the following form:

$$\dot{w} = a - bu$$

where $\dot{w}$ is the rate of wage increase, a and b are constants and u is the rate of unemployment. Phillips found that there exists a stable and inverse relationship between $\dot{w}$ and u, with the implication that lower rate of unemployment is associated with higher rate of wage increase.

Subsequent to the publication of the results by Phillips, many economists followed suit and attempted similar exercises for other countries. Subsequently, it was established that there is a stable relationship between rising wage rate and rising price level. This led some economists to refine the simple equation estimated by Phillips and use of inflation (the rate of increase in prices) instead of wage rate increase. In many cases the scatter of plot of variables appeared to be a curve, convex to origin. As empirical studies reinforced the inverse relationship between the rate of inflation and the rate of unemployment the Phillips curve soon became an important tool of policy analysis.

The policy implication of such a result was astounding – an economy cannot have both low inflation and low unemployment simultaneously. In order to contain unemployment an economy has to tolerate a higher rate of wage increase and vice versa. Thus the Phillips curve justifies the discretionary stabilization policy of a government.

In Fig. 6.1 we depict a typical Phillips curve. Suppose the economy is operating at point A with inflation rate of π_1 and unemployment rate of u_1. If the government wants to reduce the rate of inflation to u_2, the economy has to tolerate a higher rate of inflation (π_2).

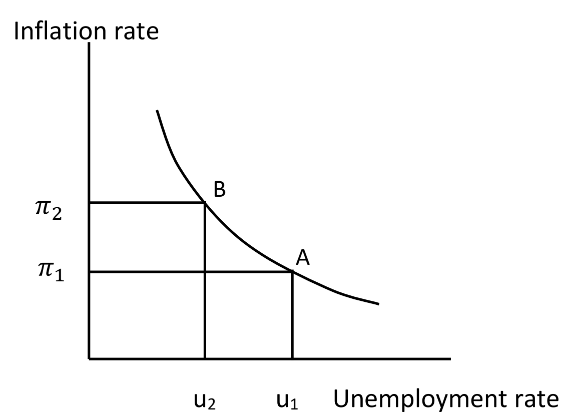

Fig. 6.1: Phillips Curve

During the 1960s and early 1970s the Phillips curve was considered to be an important tool of policy analysis. The prescription was simple and straight forward: During periods of high unemployment the government could follow an expansionary monetary policy which leaves more money in the hands of people. Such a policy may accelerate the rate of inflation while lowering unemployment. Conversely, during periods of high inflation the government could follow a contractionary economic policy so as to reduce inflation rate; the cost of such a policy however was supposed to be higher rate of unemployment. Thus, economists believed that there was a trade-off between inflation and unemployment. The government could choose any combination of inflation rate and unemployment rate depending upon the slope and position of the Phillips curve.

During the late 1970s and early 1980s, however, such a belief got shattered. The prescriptions of the Phillips curve did not work at all. Economies suffered from both high inflation and high unemployment. As unemployment increased, there was a lower level of output implying stagnation in economic growth. When governments tried to follow Keynesian policy prescription of higher government expenditure so as to increase aggregate demand, the rate of inflation accelerated. Thus, what most countries experienced was 'stagflation' – a combination of stagnation and inflation. The reason for stagflation was found to be supply shocks due the 'oil crisis' of 1973 and 1979 (refer to Unit 6). Stagflation prompted economists to explore further into the reasons for stagflation.

Check Your Progress 1

1. Write a brief note on the various types of unemployment.

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2. Explain how the Phillips curve could offer policy options before the government.

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3. Define the following concepts:
   a) Involuntary Unemployment
   b) Natural Rate of Unemployment
   c) Labour Force Participation Rate
   d) Inflation-Unemployment Trade-Off

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6.4 NATURAL RATE OF UNEMPLOYMENT

You might have come across the term full employment, which implies that all workers in the economy are employed. Have you ever thought of such a situation? Can it be attained? When we say that an economy is operating at 'full employment' level, we do not mean that there is zero unemployment. Because of imperfections in markets, rigidities in wages and prices, and various frictions in the economy it is not possible to obtain zero unemployment.

For example, at any point of time, some workers are in the transition process from one job to another (frictional unemployment). Similarly, a fraction of workers cannot be employed because of mismatch between the skill they possess and the skill required (structural unemployment).

In view of the above, a new concept termed 'natural rate of unemployment' was introduced in the 1960s independently by Milton Friedman and Edmund Phelps. Natural rate of unemployment takes into account the frictions and imperfections in the economy and assumes that it is natural for an economy to have certain fraction of its labour force unemployed, at any point of time. We observe that any unemployment that is not natural could be due to business cycle, or policy related.

For empirical purposes natural rate of unemployment is the total of frictional unemployment and structural unemployment in an economy.

It varies across countries, and over time for the same country. For the US economy, for example, natural rate of unemployment is estimated to be between 3.5 per cent and 4.5 per cent. Many countries do not report any estimate of natural rate of unemployment.

The concept of natural rate of unemployment reshaped macroeconomic analysis in subsequent years. As we will see later in this Unit, expectations of economic agents (such as households, firms and government) about future economic environment play a major role in the shape and position of the Phillips curve.

6.5 EXPECTATION-AUGMENTED PHILLIPS CURVE

The Phillips curve discussed earlier could not explain stagflation in an economy. For explaining stagflation we need to bring in expectations into our analysis. In fact, Phillips curve given in Fig. 6.1 holds true if there is no change in expectations in the minds of people. In case people perceive that there is a change in expectations, then the Phillips curve will shift. Both adaptive expectations and rational expectations have important implications for Phillips curve.

6.5.1 Phillips Curve under Adaptive Expectations

You know from microeconomics that workers and employers take decisions regarding employment on the basis of real wage; not nominal wage. According to Friedman and Phelps, expectations do matter. Thus the 'expected real wage' should be looked into account for determining equilibrium output and wage rate.

Workers usually enter into a contract with the employer regarding their salary for certain time period. During contract period, salary cannot be re-negotiated; it can be changed only after the contract period is over. As the workers are aware of these conditions, they incorporate expected inflation into the contract. For example, if the workers expect that inflation rate would be 3 per cent in the coming year, they will negotiate the wage rate in such a manner that the real wage rate does not decline due to price increase.

For an expected inflation rate of π_1 per cent, suppose the Phillips curve is given by $SRPC_1$ (see Fig. 6.2). Suppose the economy is at point A. At this point the expected inflation rate is π_1 (say 3 per cent) and unemployment rate is at the natural rate u* (say 6 per cent). At point A, the workers and firms expect an inflation rate of 3 per cent and they are getting it. Thus there is no pressure on the economy for a change.

There is a possibility of trade-off between inflation and unemployment along the curve $SRPC_1$. If there is higher inflation, then real wage will decline (because nominal wage cannot be increased due to existing contracts). Consequently, firms will employ more labour thereby leading to a decline in unemployment.

Suppose the government pursues an expansionist fiscal policy (government expenditure increases or tax rate decreases), which will boost aggregate demand. As a result, there is an increase in prices (means higher inflation rate). An expansionist monetary policy, such as increase in money supply or decrease in interest rate, will also have the same effect. It will lead to an increase in investment which will, to some extent, increase the demand for labour also. In either case, there is an increase in inflation rate to π_2 (say 6 per cent). The rate of unemployment reduces to u_2 (below the natural rate). It implies that more workers are employed, as a result of which output will be higher than potential output. In Fig. 6.2 we have shown this situation as point B.

Equilibrium at point B, however, is temporary. Workers very soon realize that there is an unexpected increase in inflation rate. In order to compensate for the price rise, workers will demand a higher wage rate. It will lead to a shift in the Phillips curve from $SRPC_1$ to $SRPC_2$. Equilibrium in the economy will be at point C in Fig. 6.2. Consequently, inflation will be at π_3 while unemployment will be at the natural rate, i.e., u*.

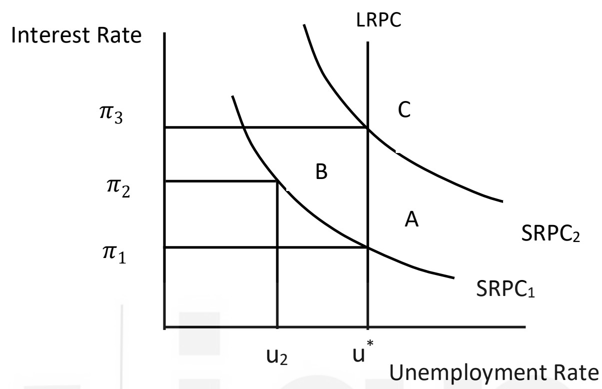

Fig. 6.2: Shift in Phillips Curve

Notice that unemployment in the economy is back at the natural rate (6 per cent). The inflation rate however is much higher (π_3). Thus the attempt of the government to reduce unemployment rate below the natural rate inflation rate, results in higher and higher inflation. In view of this, the natural rate of inflation is often termed as the 'non-accelerating inflation rate of unemployment' (NAIRU). It means that there is no acceleration in inflation if unemployment is maintained at this. Further, the term natural rate of unemployment indicates that it is inflexible, but social optimal. The term NAIRU, on the other hand, does not indicate any social optimality or desirability.

When unemployment is at the natural rate or NAIRU, there will be stability in the rate of inflation. When unemployment departs from the natural rate, there is acceleration or deceleration in inflation rate. Thus if actual unemployment is less than u^*, inflation will continue to accelerate – higher and higher in subsequent years. The concept of NAIRU and expectations formation explains the hyperinflation witnessed by many countries. Unless unemployment returns to its natural rate the inflation spiral will keep on accelerating.

The above analysis brings us to an important conclusion. Under adaptive expectations, the short run the Phillips curve is downward-sloping. In the long run however, it is vertical. In Fig. 6.2, the vertical line LRPC depicts the long run Phillips curve. Thus there is no trade-off between inflation and unemployment in the long run.

6.5.2 Phillips Curve under Rational Expectations

Under rational expectations economic agents such as firms and households are forward looking. They take into account all available information – past experience as well as present and future developments in the economy. There is no perfect foresight under rational expectations, but the errors cancel out on the whole.

An implication of the above is that actual inflation rate is equal to expected inflation rate. Thus workers and firms do not commit any error regarding wage rate during negotiations. Thus, there is no trade-off between inflation and unemployment under rational expectations. Unemployment rate in the economy is at the natural rate.

Suppose unemployment in the economy is at the natural rate. Firms and workers expect inflation to be at the rate of π_1^e. Suppose, the government pursues an expansionary policy as a result of which there is an increase in aggregate demand. Consequently, there is an increase in the rate inflation. If this policy was expected by the economic agents, they would have factored in the increase in inflation rate into their decision-making. If the policy is unexpected, then it would have the desired effect, that is, reduction in unemployment. This brings us to an important issue: how effective is government policy under rational expectations? If government policy is expected, it will not have any impact.

Check Your Progress 2

1. In 2019 the expected rate of inflation was 7 per cent while actual rate of inflation was 5 per cent. If ʎ = 0.5, find out the expected inflation rate for 2020.

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2. How do you reconcile the vertical long run Phillips curve with the downward sloping short run Phillips curve? Explain through a diagram.

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3. Explain the following concepts:

a) Adaptive Expectations

b) Rational Expectations

c) Non-Accelerating Inflation Rate of Unemployment (NAIRU)

d) Long-Run Phillips Curve

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6.6 LET US SUM UP

Unemployment results in loss of not only potential output at the macro level but also in income at the individual level. Many a time unemployment culminates into a crisis situation when there is widespread unemployment in the economy. The social sigma and psychological trauma associated with unemployment often compels policy makers to cut down on the rate of unemployment.

The classical economists assumed flexibility in real wage and prices which ensured full employment in the economy all the time. Keynesian economists, however, contest such an assumption and speak about rigidities in wage rate and prices. In case of sticky prices there is a possibility of unemployment as per the Keynesian model.

Phillips curve describes the inverse relationship between inflation and unemployment. It shows the possibility that unemployment can be reduced at the cost of higher inflation.

During the 1970s most economies in the world passed through a phase of stagflation. The trade-off between inflation and unemployment was proved to be false. In order to explain such a situation we bring in expectations into our analysis. There are two models of expectations: adaptive and rational.

According to adaptive expectations, the Phillips curve is stable in the short-run but in the long run it shifts. The long run Phillips curve is vertical. Thus there could be some trade-off between inflation and unemployment in the short-run, but in the long-run there is no trade-off. We explained the process through which the shift in the Phillips curve takes place. According to rational expectation, there is no trade-off between inflation and unemployment. Any policy of the government to reduce unemployment becomes ineffective, as people can forecast the expected changes correctly.

6.7 ANSWER/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1. Your answer should include frictional, structural and cyclical unemployment. Go through Section 6.2 for details.
2. Go through Section 6.3 and refer to Fig. 6.1.
3. These concepts are discussed in Sections 6.2 and 6.3.

Check Your Progress 2

1. Refer to the text in Section 6.5. You should explain Fig. 6.2.
2. These concepts are defined in Sections 6.5.

UNIT 7 RATIONAL EXPECTATIONS *

Structure

7.0 OBJECTIVES

After going through this Unit, you should be in a position to

- explain the concept of rational expectations and differentiate it from adaptive expectations;
- explain the relationship between the AS-AD model and the non-neutrality of money;
- discuss the Lucas critique;
- bring out the relationship between perfect foresight and neutrality of money;
- describe the rational expectations hypothesis;
- discuss the Lucas supply function; and
- explain the policy ineffectiveness theorem.

7.1 INTRODUCTION

Several important decisions by economic agents are based on expected values of economic variables. Let us take, for example, the case of Friedman's permanent income hypothesis. According to Friedman consumption expenditure depends not on current income, but on permanent income. Thus a reduction in income tax will

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lead to an increase in consumption expenditure only if the reduction in tax rate is expected to be permanent; not temporary. Similarly, how much labour a worker will supply to the market is based on his/her expectations of future real wages. While nominal wages are negotiated before the work is undertaken, they are received afterwards. Thus, real wages (= nominal wages/ aggregate price level) received by labour is known only in the future, when the worker goes to spend it at the prevailing price level of goods and services. An implication of the above is that how the future expected price level is determined is critical in decision-making. Expected price level determines expected real wages; it further determines whether a worker will continue to supply labour effort or withdraw it. Therefore, it is important to know the way expectations are formed.

Recall from Unit 1 that the classical economists assumed money to be neutral. Its implication is that a change in money supply affects nominal variables (e.g., price level) only while real variables (e.g., output, employment) remain unaffected. Most neo-classical economists, using the AS-AD approach, however realize that money does have an impact on real variables such as real wages, unemployment and output in the short run. This short-run impact of money arises due to the fact that there is a lag between expected value of real wages and actual value of real wages.

The initial impact of an increase in money supply is to increase the demand for goods and prices. Workers offer more labour, being 'fooled' into thinking that their real wages have increased, while actually real wages have not. In these 'fooling models', firms do not have any illusion – the firms realize that real wages have decreased, and therefore the firms hire more workers.

In this Unit we will discuss the non-neutrality of money in the short run using the AS-AD approach. Then, we will move on to the Lucas Critique which explains why agents are unlikely to form expectations that will persistently underestimate price rises. We will illustrate the results of the AS-AD model and contrast it with the case of 'perfect foresight', where neutrality of money would hold even in the short run.

We will discuss how expectations are formed using 'rational expectations'. As the name suggests, rational expectations refers to the value that a variable will take in the future. Rational expectations are formed on the assumption of rationality, i.e., based on incorporating all information about the variable and what its factors, i.e., knowing the true model explaining the variable. Then the variable is totally predictable except for random factors that cannot ever be anticipated. Uncertainty about the impact of random events or lack of global information can lead to less than 'perfect foresight', i.e., the variable not being totally predictable, or the expected value differing from the actual value in the future.

Rational expectations models come under the purview of new-classical economics. We know that in classical economics, there is perfect information, perfect flexibility in wages and prices and instantaneous market-clearing. These features of classical economics ensure full employment in the labour market and corresponding full employment output. The new-classical economists allow for imperfect information (agents may have local information, not global information). Thus, workers and firms may not know the aggregate price level, although they would know their own wages and price levels. Unlike the AS-AD model, rational expectations models are not based on 'asymmetric information' across agents – workers and firms have access to the same information and are equally 'fooled' by unanticipated events. We will talk about this in details while discussing the Lucas Supply Curve. Finally, we will discuss about the 'policy ineffectiveness proposition'. This proposition uses the Lucas Supply Curve and the assumption of rational expectations to show that anticipated monetary policy is neutral with no effect on output and that only the unanticipated part of monetary policy impacts output. Thus it makes ineffective the use of any systematic monetary policy.

7.2 THE AS-AD MODEL AND NEUTRALITY OF MONEY

The aggregate demand (AD) function shows combinations of the aggregate price level (P) and real income (Y) which show equilibrium in the goods and money markets. It is a downward sloping curve reflecting that lower price levels are associated with higher real money balances and the resulting lower interest rates are associated with higher investment expenditure and therefore higher aggregate demand.

The AD function: $Y = f(G, T, M/P)$ is an increasing function of G, T and M/P, where G= Government expenditure, T= Taxes; M/P = nominal money supply/price level.

The aggregate supply (AS) equation shows labour market equilibrium – higher levels of output are associated with higher employment, higher wages due to the tighter labour market and, with cost plus pricing, imply higher prices.

The AS function comes from

a) the wage function: $W = P^e f(u,z)$

   where W= wage rate; P^e = expected price, u = unemployment rate , and z is a variable capturing all other factors affecting W

b) the price setting relationship: $P = (1 + m)W$, where m = mark-up over cost (lower for a more competitive economy).

Hence, the AS function is $P = P^e (1 + m) f(u, z)$

Or, $P = P^e (1 + m) f(1 – Y/L, z)$

where u is replaced by $(1 – Y)/L$ (assuming a simple production function Y = N = number of employed out of the total labour force = L).

When $P = P^e$, the real wage is what firms and workers have to jointly determine in the labour market (given *u* and *z*), and the resulting rate of unemployment is the 'natural rate of unemployment'

Now $P = P^e$ coincide in the long run as any gap between price and expected price decreases with workers reducing the expectation error over time. So the natural rate of unemployment prevails in the long run.

However, in the short run, any divergence between actual P and expected P implies that workers' expected real wage level is not met and the AS curve will shift accordingly, leading to output levels changing from the 'natural level of output'.

In Fig. 7.1, we start with aggregate demand curve (AD_0) and aggregate supply curve (AS_0). Output is at the natural level of output Y_N and price P_0 coincides with the expected price P_0.

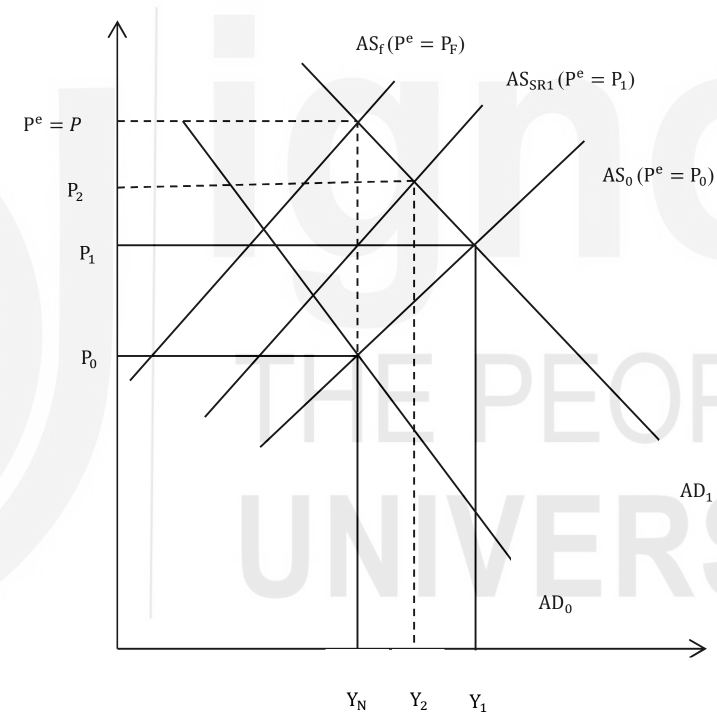

Fig. 7.1: The AS-AD Model without Rational Expectations

Let us assume that there is an increase in money supply. This will shift the AD curve to the right (from AD_0 to AD_1) and result in higher output Y_1 and a higher aggregate price level P_1 (intersection of the AS and AD curves). Thus, in the short run (where P^e is fixed), the increase in the money supply has caused output to change and we see that money is not neutral. This short run impact of money comes from the lack of anticipation by workers on the future impact of the increased money supply on prices and real wages, i.e., due to the miss-match of P and P^e. The initial impact of the increase in money supply is to increase demand for goods, demand by firms for more labour and nominal wages. However, workers do not anticipate that prices will also rise and that while nominal wages have risen, real wages have fallen since P rises more than P^e (which initially is constant and on what the AS curve depends). Therefore, workers offer more labour effort, being 'fooled' into thinking that their real wages have increased.

The traditional Phillip's Curve showing the trade off between price rise and unemployment was based on the same phenomenon of assuming that price expectations of workers are unchanged even when a price rise is observed.

Only when workers spend their wages and realize that the price level (P_1) is higher than they had expected (P_0), they realize that the real wages have actually declined. Then the new wage contracts will demand higher nominal wages, and the AS curve shifts up to AS_2. Here again, actual prices are higher than the expected prices P_1, output is lower at Y_2 but still higher than Y_N. The economy is still in the short run. The process continues iteratively, till the AS curve is finally at AS_f, where $P = P^e$ again, workers are getting their desired real wage, and output is back at Y_N.

This is the same as what the 'modified' Phillip's Curve shows: price expectations rise when price is greater than expected price, so only a price rise higher than expected will lead to an increase in employment or output, Y. Notice that in these models, firms unlike workers realize that real wages have declined and therefore hire more workers. Thus, there is asymmetric information on the part of workers and firms.

In the above model, workers' expectations of the price level are based on the past period's price level. In other words, expectations of workers is $P_t=P_{t-1}$. This is a form of adaptive expectations where expectations are changed according to past observed actual values.

7.3 THE LUCAS CRITIQUE

The concept of rational expectations was introduced by John Muth in 1961. Robert E. Lucas and Thomas Sargent introduced this concept in economics through their writings in 1970s.

In the AS-AD model discussed earlier, money is neutral in the long run (when $P=P^e$). In the short run, however, workers think that their nominal wage increase is a real wage increase. Thus, in the short run, money can have real impact on employment and output.

Suppose, the policy maker increases the growth rate of money supply, as a result of which there is a price rise. The expected price is P_0, but actual price turns out to be P_1. Economic agents adapt their expected price to be P_1 in the next period. Actual price however increase further to P_2. So, even though workers do not realize it (since their expectations of price are taking time to catch up to the actual price), real wages will fall – there is short run impact of money.

Lucas questions this 'fooling' of workers after past monetary policy has been used to increase employment. Having observed similar episodes of rising prices after increases in money supply in the past, should not economic agents raise their expected price? They should take lessons from the past along with all relevant information in order to make rational predictions. A sensible forecast of price should be made on the basis of the behavior of the actual economy i.e. understanding the process or true model that generates the actual price and any new relevant information that comes up.

Thus, Lucas questioned the premise that people's expectations would remain unchanged when policy makers take advantage of a relationship based on past data on employment and inflation. The past relationship was based on people's average expectation of inflation remaining unchanged at zero. If policy makers have been increasing money supply to increase employment, resulting in a persistent price increase, people's expectations of inflation are now likely to increase, with the result that the increased money supply leads to higher prices but not higher employment.

This is the 'Lucas critique' – past statistical relationships in data are based on a certain expectations, when the policy environment changes, expectations are likely to change and the statistical relationships displayed in the past may no longer be hold.

As another example, let us take the case of policy makers decreasing the growth of money supply. The objective of the policy maker is to reduce inflation. Suppose, people base their expectation of future prices on last year's higher prices. Actual prices, however, would be lower due to decline in money supply growth rate. This will result in higher real wages and lower employment.

In the above example, suppose the government announces that it would decrease inflation by reducing the growth of money supply (and people believe the government because of its past records). Consequently, people with rational expectations will incorporate this relevant new information, and reduce their expectations of the price level. Accordingly, the nominal wages would decrease. Therefore, when actual prices fall, there is no increase in real wage and unemployment.

According to Lucas, people make rational forecasts on variables (such as prices) on the basis of policy environment and available information. Lucas' critique is that when there is a change in the policy environment, economic agents will revise their expectations.

7.4 PERFECT FORESIGHT AND THE NEUTRALITY OF MONEY

There is the 'perfect foresight' if we know the AS-AD model equations and parameters that determine the actual price, P. If we use this model to get the value of P^e (expected price), the P^e should equal P. That means there will be no expectational error ($P - P^e = 0$). Consequently, workers would not be fooled into thinking that an increase in nominal wages is an increase in real wages.

In that case, there would be no increase in employment that would result from the increase in money supply and money would be neutral. Note that the above also requires that wages and prices are totally flexible, so that the real wage can be maintained at the desired level at all times and employment remains at the natural rate of unemployment. Thus, it is very similar to the Classical Model with regards to its performance.

In Fig. 7.1, if $P^e = P$, then the level of output would remain at Y_N. In the case of perfect foresight, price would increase proportionate to the increase in money supply immediately, real wages would remain unchanged and there is no change in employment level. There would be no role for monetary policy to play in this case because every policy that would be implemented would be immediately understood by the public. As a result, only nominal variables would be affected and real variables will remain unchanged.

However, in the real world, especially for economic variables which are impacted by human behaviour with random elements, there is no 'perfect foresight'. At best, people can make 'rational expectations' on which to base decisions.

Check Your Progress 1

1. Describe the AS-AD model. Illustrate the condition of natural rate of unemployment in a diagram.

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2. Illustrate the impact of an increase in money supply in the short-run through the AS-AD diagram.

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3. Explain the Lucas critique on the effectiveness of monetary policy.

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4. What is meant by perfect foresight? Under such condition, how do the changes in money supply affect an economy?

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7.5 RATIONAL EXPECTATIONS HYPOTHESIS

In this section we will discuss as to how rational expectations are formed. Also we will discuss the properties and implications of rational expectations hypothesis (REH). We will start with a model that depicts 'perfect foresight' and then contrast it with a rational expectations model.

The future value of a variable, e.g., prices, interest rates or output, can be predicted or guessed in several ways – through astrology, throwing darts, based on past trends, adapting (changing) or correcting past expectations using more recent data on the variable. However, the above approaches are unlikely to generate good predictions even on average because they ignore the underlying process that influences the variable.

The method of rational expectations assumes that individuals have observed enough history on the concerned variable to understand its explanatory factors, and the systematic manner in which these explanatory factors influence the variable. Thus, individuals use their knowledge about the process of generating the variable. These systematic patterns then restrict the values that the variable can take. This helps individuals in forming their expectations about the likely future value (or expected value) of the concerned variable.

As an example, using a general algebraic representation, we assume that the 'true model' (process that generates variable *Y*) is known with the relevant variables affecting it are *X, Y* and *Z*. The relationship between the variables given as follows:

$$Y_t = \alpha + \beta X_{t-1} + \gamma Y_{t-1} + \eta Z_{t-1} \qquad \dots (7.1)$$

where Y_t is the actual value of Y at time *t* and is a function of observed values at time (t-1) of variables X, Y and Z. Assume that α is a constant term, and β, γ and η are coefficients of observed values of X, Y and Z.

Equation (7.1) is a deterministic model where the right hand side variables completely determine the value of Y. Now, the mathematical expectation of Y_t is:

$$E_{t-1}Y_t = \alpha + \beta X_{t-1} + \gamma Y_{t-1} + \eta Z_{t-1} \qquad \dots (7.2)$$

where $E_{t-1}Y_t$ is the expected or predicted value of Y made at the end of time period (t-1) for the future time period *t* based on observed values at time (t-1) of variables X, Y and Z.

Equation (7.1) is also a model where agents have 'perfect foresight' and can accurately predict the value of Y. Thus, expectational error = [actual value of Y (that is, Y_t) – expected value of Y (that is, Y^e_t)] = 0 in the case of equation (7.1). As an example, with perfect foresight, the expected price level following an increase in the money supply would be an equi-proportionate increase in the price level.

Let us try to re-formulate equation (7.1) in terms of rational expectations. The actual value of Y in period *t* depends not only on the variables X, Y and Z and coefficients α, β, γ and η, but also on a random term ν_t. The random term ν_t reflects the inherent stochastic (probabilistic) process in an economy (for example, the unpredictable nature of human behaviour). Thus,

$$Y_t = \alpha + \beta X_{t-1} + \gamma Y_{t-1} + \eta Z_{t-1} + \nu_t \qquad \ldots (7.3)$$

Now, $E_{t-1}Y_t$ below is the expected or predicted value of Y made at end of time period (t-1) for the future time period *t* based on observed values at time (t-1) of variables X, Y and Z.

$$E_{t-1}Y_t = \alpha + \beta X_{t-1} + \gamma Y_{t-1} + \eta Z_{t-1} \qquad \ldots (7.4)$$

This expected value above is likely to happen or realize on average. The actual value of Y in period t, on the other hand, can be quite different depending on the random element ν_t. Usually, the value of the stochastic error term is small, as it is a composite of several variables; different variables pulling Y_t away from its average value in different directions. Thus, the final value of Y_t is never totally predictable since the random term is knowable only at time *t* and can be thought of as the effect of human behaviour which is individual, unique and not entirely predictable.

The random error term $\nu_t = (Y_t - E_{t-1}Y_t)$ is also the expectational error of the rational expectations prediction. We can say that the distribution of ν_t has mean zero and known variance σ^2. Thus, on average, the expectational error will be zero (because if not, we can change the value of α to make it so). The rational expectations value will also be the most accurate expectations since it is based on the true model on which the concerned variable is based.

Let us consider an example: a simple model of consumption may have C_t depending on past wealth, past consumption and past disposable income so the rational expectations value of consumption will be the average or expected consumption level of a set of people. However, each individual's future consumption level will not and cannot be exactly predictable – there will be some variation around the average or expected consumption due to different preferences, mostly close to the average.

According to the rational expectations hypothesis, economic agents in the model (workers, consumers, firms, the Government) are knowledgeable about the true model. They have learnt through observation the behaviour of all the relevant variables (X, Y, and Z) and parameters (α, β, γ and η) in explaining Y. They know the functional form (linear here; could be quadratic, exponential or some other type) of the relationship. Thus, an individual with rational expectations can predict everything predictable about the future values of Y. The only expectational error to emerge then will be due to the inherently unpredictable portion of the variable.

Rational expectations hypothesis can be criticized for its assumption of rationality and availability of full information. People, in general, are not aware of the economic processes that take place in an economy. To assume that people can figure out the explanatory variables of the concerned variable and that they have all relevant information is unrealistic. However, people these days have access to readily available forecasts by experts via multiple public media. Certain relevant information may be expensive to procure. The rational expectations prediction will still be correct on the average, though it may not be as accurate as it could have been with the variable.

Despite these criticisms, to use all relevant available information to form rational expectations when making decisions is nevertheless an appealing underlying microfoundation for macroeconomic theory. It assumes that individual agents optimize and make decisions that are rational and most beneficial to them. It also leads to the outcome that with perfect information and rational expectations there is market- clearing. This is due to the fact that each agent (firms and workers) is making decisions. Firms decide on the quantity of output to produce on the basis of a production function while workers decide on the amount of labour to supply on the basis of a utility function. Notice that microfoundation is an integral part of rational expectations hypothesis. With flexible prices and wages, demand and supply for output and labour will match each other in their respective markets. Thus, there will be no involuntary unemployment or no deviation of output from the natural level of output.

We need to see whether the outcomes as predicted by rational expectations at the micro level can explain macroeconomic relationships, e.g., between unemployment and inflation. One key implication of rational expectations is that government policy to reduce unemployment (monetary expansion or monetarization of debt due to fiscal expansion) will not have the desired effect if it is predictable or announced in advance. Monetary expansion to increase aggregate demand and raise output, will not lead to higher employment if workers with rational expectations anticipate that all prices (nominal wages and prices of what they buy) will increase proportionately and that their real wages are not going to increase.

A question arises at this point: How would rational expectations explain observed booms and recessions? During a boom (recession) unemployment rate decreases (increases), while actual output deviates from natural output. We will discuss these issues in the next sections while discussing the 'Lucas supply function' and the 'policy ineffectiveness proposition'.

7.6 LUCAS SUPPLY FUNCTION

According to Lucas, economic agents care about their relative price, i.e., about the price that they will receive for what they sell, relative to the price that they will pay for what they buy. Thus a firm cares about the price of the output it is selling, relative to the price of inputs that it buys, while a worker cares about the real wage rate, i.e., the nominal wage, which is the price for the labour services that he/she sells versus the prices of the goods that he/she will buy. An increase in the firm's relative price will then bring forth an increase in the output of firms while an increase in the real wage will elicit higher labour supply (assuming the substitution effect is greater than the income effect in the labour supply decision).

However, Lucas also postulates that agents are unlikely to have perfect information on their relative prices because they know much more about the price of what they are selling than about the goods or services they will buy since there are several inputs/ goods that they will buy relative to what they specialize in and sell. Thus, firms and households have local information and not global information. This can 'fool' them into making incorrect decisions (from their welfare point of view). Suppose, there is an increase in the price of the good they sell (or labour service), then each agent (firm/ worker) needs to determine whether the increase is only or mainly in the price of their good/ labour or there is a universal increase in prices of all the goods. Each agent thus has a 'signal extraction problem' to distinguish between a relative prices increase vs. an increase in the aggregate price level.

Lucas further postulates that economic agents face a second signal extraction problem. Temporary increases in relative prices are more important than permanent increases. Firms and workers need to be quick in taking advantage of profitable opportunities. If in the past, most increases in selling prices have been partly increases due to general inflation and partly due to relative price increase, then both firms and workers observing a general price increase will assume that some of the price increase is a relative price increase for their product/ labour service and will increase their supply of it.

Now, if there is general increase in the price level, say due to an increase in the money supply, and all agents knew this, with rational expectations, they would understand that all prices had increased proportionately and there is no relative prices change, then there would be no change in labour supply or in output. Thus any anticipated, expected or announced change in the aggregate price level will not lead to any change in output. However, if there is a 'price surprise', i.e. a difference between actual price level and the expected price level, then output (or labour supply for workers) will increase.

We can thus write the Lucas Supply Curve as:

$$Y_t/Y_p = f(P_t/E_{t-1}P_t) \qquad \text{... (7.5)}$$

where Y_t = output in period t;

Y_p = potential output (i.e., full employment output);

P_t = price of product; and

$E_{t-1}P_t$ = price of product in period *t* expected one period prior, i.e., in period (t-1).

In logarithmic terms, we can write equation (7.5) as

$$y_t - y_p = f(p_t - E_{t-1}p_t) \qquad \text{... (7.6)}$$

where the variables in lower case indicate the variables in logarithmic term.

Equation (7.6) indicates that output is higher than the full employment (or natural level of) output when there is a positive price surprise reflecting a relative price increase for the product (or labour service) being sold.

In an environment where prices have been stable, and most price increases have meant a relative price increase for the product/ service being sold by the firm/worker, both firms and workers can be 'fooled' into supplying more output/labour supply if the money supply is increased suddenly. On the other hand, if the prices have generally been volatile (as in some Latin American economies with high inflation), monetary expansion will most likely have no impact on output or labour supply since agents will not be fooled into thinking that a price increase is for their product alone, but will realize that the price increase is for all products due to general inflation.

If there was perfect information so that people could distinguish between general (i.e., aggregate) price increase vs. a relative price increase. In other words, if expected price increase mirrored the actual price increase, there would be no increase in output for firms or increase in labour supply for workers. But since both firms and workers get 'fooled' into thinking that *their* price has increased (because they only have local information on their sale price and not global information on all prices), that all firms and workers tend to increase output and labour supply. According to Lucas, such 'fooling' behaviour leads to fluctuations in the level of output. This explains business cycle in an economy. Note that unlike the AS-AD approach of Section 7.2 where firms knew the correct real wages but workers did not, in the Lucas Supply Curve both the agents (firms and workers) are symmetrically 'fooled',

The Lucas supply function relies on a price surprise to explain why output fluctuates from the 'full employment' or 'natural level of output' level. Had there been perfect information, so that prices could be anticipated accurately, agents would be able to distinguish between increases in nominal vs. real price increases and we would get the results of the classical economists i.e., dichotomy between the real and monetary sectors of the economy and that money is neutral.

7.7 POLICY INEFFECTIVENESS THEOREM

While the Lucas supply function focused on the supply side, the policy ineffectiveness theorem focuses on the demand side to explain deviations of output from the natural level of output. Here again, these deviations of output and employment are explained by price surprises caused by the non-anticipatable portion of money change. The model here is based on the seminal paper "Rational expectations and the theory of economic policy" by Sargent and Wallace (1976) [see, Sheffrin (1996)]. Note that only monetary and no fiscal changes are being considered here.

Let us assume that economic agents have rational expectations. Therefore, they have learnt from past observation how the monetary authorities operate, i.e., what is their money supply rule (see Unit 18 for further discussion on this). However, there is a random, unpredictable portion to the money supply since it is also dependent on factors beyond the central bank's control (e.g., capital flows in and out of the country; banks may keep reserves higher than the required reserve ratio; public may decide to hold more cash than bank deposits). According to the 'policy ineffectiveness theorem', output will not deviate (from the full employment level) with a change in money supply unless it is unpredictable. Any predictable change in money supply will lead to a change in expected prices such that there will be a corresponding change in actual prices, but no change in output. Prices, on the other hand, will be influenced by both predictable and unpredictable parts of any money supply change. The model consists of an aggregate demand relation, aggregate supply relation and a money supply rule, along with an equation capturing rational expectations:

1) Aggregate Demand Relation:

$$M_t + V_t = P_t + y_t \quad \ldots (7.7)$$

Equation (7.7) is the familiar MV=PY equation of classical theory except that each term above is in log form and therefore is a linear equation.

M_t = log of the money supply

V_t = log of the (assumed constant) velocity of money

P_t = log of the price level

y_t = log of real output

Allowing V_t to vary with interest rate is not critical to the argument so the assumption that it is constant is for simplification.

This can be seen as the AD line in Fig. 7.2. An increase in M_t will shift the AD line to the right.

2) Aggregate Supply Relation:

$$y_t = y_p + \beta(P_t - {}_{t-1}P^e_t) \quad \ldots (7.8)$$

Where

y_p = log of full employment output

${}_{t-1}P^e_t$ = log of the price level expected in period *t*, with the expectation formed in period (t-1)

This is based on the Lucas Supply function. Output will differ from the full employment level of output if there is a price surprise, i.e., the price level in period *t* was not anticipated. In Fg. 7.2, this is the AS line. It shows that $y_t = y_p$ if $P_t = {}_{t-1}P^e_t$. However, y_t increases relative to the fixed y_p if P_t is larger than the expected ${}_{t-1}P^e_t$ (that is, when there is a positive price surprise).

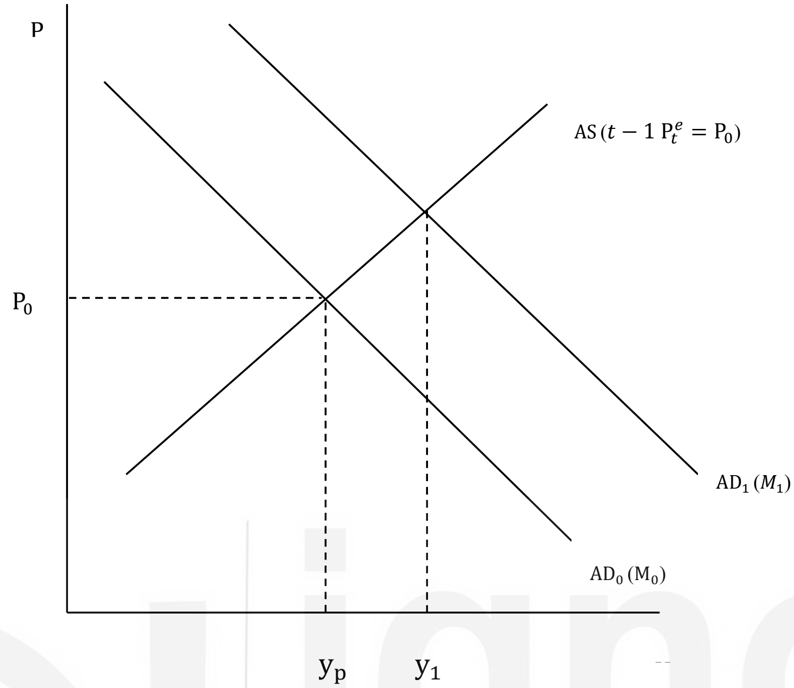

Fig. 7.2: Lucas AS Curve

3) Money Supply Rule:

$$M_t = \alpha y_{t-1} + \epsilon_t \qquad \ldots (7.9)$$

where $E(\epsilon_t|I_{t-1}) = 0$

Implication of equation (7.9) is as follows: The monetary rule has a portion αy_{t-1} that is under the control of the monetary authorities. This portion is based on past observed data which is known to the general public as well and can be anticipated by them. However, there is a random component, ϵ_t in equation (7.9) which nobody can know before the end of period *t* and cannot be predicted. All we can say is that the expected value of this random component is zero given the information $I_{(t-1)}$ (available at the end of period t-1) on the basis of which any prediction for period *t* can be made.

4) Rational Expectations:

$$_{t-1}P^e_t = E(P_t|I_{t-1}) \qquad \ldots (7.10)$$

Equation (7.10) captures the following: With rational expectations, the expected value of price in period *t* (which is based on information I available in period (t-1)) is endogenous. Endogenous means that the variable (in this case, prices) is derived within the model. Thus, the same model that explains prices is used to estimate expected prices. The price level in period *t* that people 'feel' will prevail or psychologically expect (that is, $_{t-1}P^e_t$) is obtained from the model as the mathematical expectation of prices in period t based on the information available in period (t-1), that is, $E(P_t|I_{t-1})$.

Before we move to the mathematical derivation of the policy ineffectiveness theorem, let us graphically see its results. In Fig. 7.2, if money supply is increased (based on the money supply rule given at equation (7.9)), the AD curve AD_0 ($P_0 = M_0 + V - y_p$) will shift to AD_1 ($P_t = M_t + V - y_t$). In the absence of any change in expected future prices, the AS curve (drawn for a particular expected price) will remain unchanged, and the shift in the AD curve will result in increased output as well as prices, from y_p to y_1. This is consistent with the original Phillip's Curve. It shows a trade-off between lower unemployment (increased output) and price rise.

Given rational expectations, in period (t-1) itself, people will anticipate the shift of the AD curve due to the anticipated or expected component (not the random portion) of the money supply rule. Therefore, the AS curve will shift up corresponding to the price increase expected due to this expected component. Such as shift in the AS curve is shown in Fig. 7.3 with the dotted AD curve (which is only the anticipated and not the actual increase in the price level).

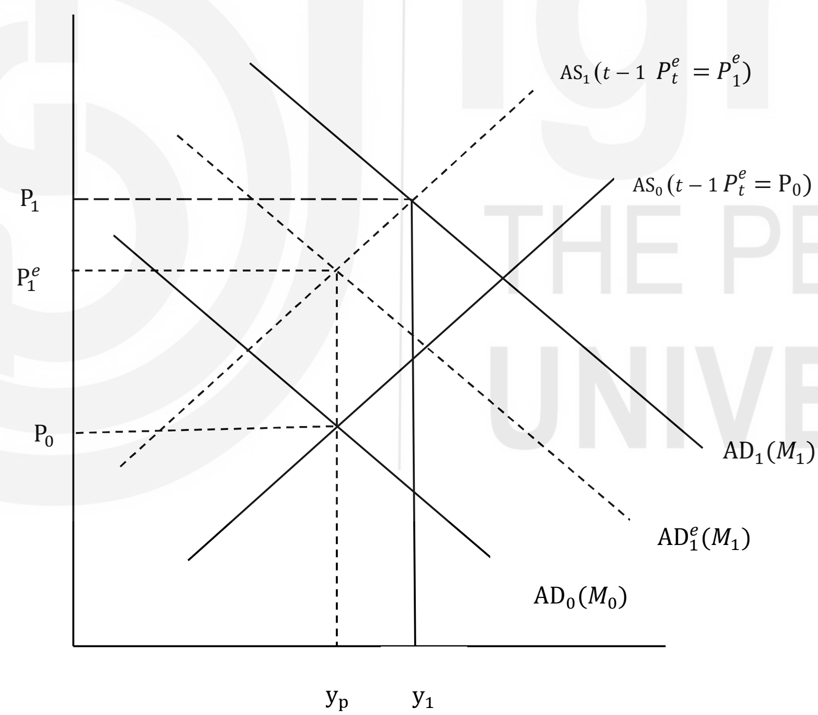

Fig. 7.3: AS-AD Model with Rational Expectations

If the random term ϵ_t in the money supply rule were zero, the actual price level in period t would be exactly as anticipated; there would be no price surprise in the Lucas supply function. Therefore, there is no change in the output level. However, actual money supply M_t can be different from the expected money supply M^e_t due to the random term ϵ_t. In Fig. 7.3, we have shown the difference to be positive, as a result of which AD_1 is to the right of the dotted AD^e_1. The actual price level can be different from the expected price level ${}_{t-1}P^e_t$ and due to this price surprise, the actual output will differ from the full employment level of output y_t.

Thus, both the anticipated and unanticipated portions of the increase in money supply result in an increase in the price level. However, only the unanticipated portion of the money supply increase which shifts the AD curve unexpectedly leads to the price surprise, and therefore to a change in the output level.

Now, let us combine the four equations (7.7), (7.8), (7.9) and (7.10) together.

1) Aggregate Demand Relation: $M_t + V = P_t + y_t$ ...(7.7)

2) The Aggregate Supply Relation: $y_t = y_p + \beta(P_t - {}_{t-1}P^e_t)$ ...(7.8)

3) The money supply rule: $M_t = \alpha y_{t-1} + \epsilon_t$ ...(7.9)

4) Rational Expectations: ${}_{t-1}P^e_t = E(P_t|I_{t-1})$ ...(7.10)

Let us substitute y_t from (7.8) and M_t from (7.9) in (7.7). We get

$$\alpha y_{t-1} + \epsilon_t + V = P_t + y_p + \beta(P_t - {}_{t-1}P^e_t) \quad \text{... (7.11)}$$

Now let us take the mathematical expectation of the above equation as of time (t-1) to get ${}_{t-1}P^e_t$

$$\alpha y_{t-1} + E(\epsilon_t) + V = {}_{t-1}P^e_t + y_p + \beta({}_{t-1}P^e_t - {}_{t-1}P^e_t).$$

This simplifies to

$\alpha y_{t-1} + V = {}_{t-1}P^e_t + y_p$ or that

$${}_{t-1}P^e_t = \alpha y_{t-1} + V - y_p \quad \text{… (7.12)}$$

From (7.7) we know that $P_t = M_t + V_t - y_t$. Substituting for M_t from (7.9), we get

$$P_t = \alpha y_{t-1} + \epsilon_t + V - y_p + \beta(P_t - {}_{t-1}P^e_t) \quad \text{… (7.13)}$$

Subtracting ${}_{t-1}P^e_t$ in equation (7.12) from P_t in equation (7.13), we obtain:

$$P_t - {}_{t-1}P^e_t = \epsilon_t + \beta(P_t - {}_{t-1}P^e_t) \quad \text{… (7.14)}$$

Equation (7.14) can be re-arranged to show that the gap between the actual and expected price level is

$$P_t - {}_{t-1}P^e_t = \epsilon_t/(1+\beta) \quad \text{… (7.15)}$$

From the Lucas Supply Function given at equation (7.8) we find that

$$y_t = y_p + \beta(P_t - {}_{t-1}P^e_t)$$

which means

$$y_t = y_p + \beta\, \epsilon_t/(1+\beta) \quad \text{… (7.16)}$$

Equation (7.16) implies that output (y_t) is higher than full employment output (y_p) only if there is some unexpected or unanticipated increase in the price level. From (7.15), we find that $P_t = {}_{t-1}P^e_t + \epsilon_t/(1+\beta)$. It means that both the expected price level and the unanticipated portion of the money supply affect the price level. These are the two results we set out to show in the policy ineffectiveness proposition.

Only unanticipated (or, surprise) changes in monetary policy can influence output, given rational expectations and thereby knowledge of how the monetary authorities react to a given situation through the money supply rule. This money supply rule could have taken a different functional form, but to the extent that it is known, it cannot have any effect since the public anticipates it and changes expectations accordingly. The policy ineffectiveness proposition thus negates the possibility of any systematic policy rule to keep the actual output level of the economy above its potential output level.

Both the Lucas Supply Curve and the Policy Ineffectiveness Proposition indicate that money is not neutral in the short run. This is due to the fact that unanticipated high prices fool the economic agents to voluntarily work harder or increase output. Keynesian economists criticize this approach by arguing that correct information should be available within a short time, so that booms and recessions spanning several years are not well explained by rational expectations. They also criticize the assumption that all markets clear (which implies that people are voluntarily unemployed) when there is a high rate of unemployment.

Despite the criticism, we can say that rational expectations is an appealing way to incorporate microeconomic foundations of macroeconomic relations and variables. This has several interesting implications. In consumption, for example, the random walk of consumption and the issue of Ricardian Equivalence.

Check Your Progress 2

1) Bring out the salient features of the Rational Expectation Hypothesis. Explain it by considering three variables.

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2) Describe the Lucas supply curve.

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3) What is meant by policy infectiveness theorem?

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7.8 LET US SUM UP

In this unit we learnt about the relation between price and unemployment, the short-run and long-run impact of money supply on output and prices by discussing the non-neutrality of money, traditional and modern Phillips curve, and the adaptive expectations. We also learnt about the Lucas critique and the perfect foresight as related to the effect of monetary policy in an economy. In the perfect foresight model, we learnt that the prediction is accurate by which the expected error is zero. However, an economic variable may be dependent on a random component that reflects the inherent stochastic process of the variable, on which the rational expectations hypothesis is based on. Then we discussed the formation of rational expectations hypothesis, its properties and implications. We also learnt about the Lucas Supply function, which focused on the supply side. In contrast, the policy ineffectiveness proposition focuses on the demand side to explain deviations of output from the natural level of output that are explained by price surprises caused by the non-anticipatable portion of money change.

7.9 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) The AD function: $Y = f(G, T, M/P)$ is an increasing function of G, T and M/P, where G= Government expenditure, T= Taxes; M/P = nominal money/price level. The AS function is $P = P^e (1 + m)f(u, z)$. Use Fig. 7.1. Refer to Section 7.2.

2) The traditional Phillip's Curve shows the trade off between price rise and unemployment, which is based on the same phenomenon of assuming that price expectations of workers are unchanged even when a price rise is observed. The 'modified' Phillip's Curve shows that price expectations rise when price is greater than expected price, so only a price rise higher than expected will lead to an increase in employment or output, Y. Use Fig. 7.1.

3) Refer to Section 7.3 and answer.

4) There is the 'perfect foresight' if the true AS-AD model equations and parameters are known. It implies that people have perfect information on the true model that generates actual P. If the same model is used to get the value of P^e (expected price), then P^e should equal P. That means there will be no expectational error (that is, $P – P^e = 0$).

Check Your Progress 2

Check Your Progress 2

1) Refer to Section 7.5 and answer.

2) The Lucas supply curve relies on a price surprise to explain why output deviates from the natural level output. In the absence of perfect information economic agents are not in a position to anticipate future prices and wage rates accurately. They cannot distinguish between increases in nominal vs. real prices due to signal extraction problem. Thus, there is no dichotomy of real and nominal sectors, and money is not neutral in the short run. Refer to Section 7.6.

3) The policy ineffectiveness theorem focuses on the demand side to explain deviations of output from the natural level of output, which is explained by price surprises caused by the non-anticipatable portion of money change. The model consists of an aggregate demand relation, aggregate supply relation and a money supply rule, along with an equation capturing rational expectations. Use Fig. 7.2 and Fig. 7.3 to show the effect of increasing money supply. Use equations (7.7), (7.8), (7.9) and (7.10) to derive equations (7.15) and (7.16). Refer to Section 7.7.